

CUSTOMER SEGMENTATION PROJECT REPORT

Objective

The objective of this project is to identify distinct customer groups based on demographic and behavioral characteristics. Customer segmentation helps businesses understand purchasing behavior and develop targeted marketing strategies.

Tools Used

- Python (Pandas, Scikit-learn, Matplotlib)
- Power BI
- CSV Dataset

Methodology

The dataset includes customer age, gender, annual income, spending score, and purchase frequency. Data was preprocessed and standardized before applying the K-Means clustering algorithm. Four customer segments were created based on similarities in income and spending behavior.

Key Findings

1. High-income and high-spending customers represent the most valuable customer segment and contribute significantly to revenue.
2. High-income but low-spending customers indicate potential opportunities for targeted promotions and engagement campaigns.
3. Low-income but high-spending customers demonstrate strong purchasing intent despite budget constraints.
4. Low-income and low-spending customers show limited purchasing activity and may require different marketing strategies.
5. Spending behavior varies significantly across customer groups, highlighting the importance of personalized customer engagement.

Conclusion

Customer segmentation provides actionable insights into customer behavior and preferences. The identified segments can help businesses optimize marketing campaigns, improve customer retention, and increase overall profitability through targeted decision-making.